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Frequency (commonness, probability, likelihood)

$$\omega = \frac{d}{dt}\varphi = \eta kv = kc = \frac{2\pi}{\mathcal{T}} = 2\pi \cdot \nu = \frac{a}{v} = \frac{v}{r} = \frac{E}{\hbar} = \sqrt{\frac{k}{m}} = \sqrt{\frac{r \times F}{m}} = \sqrt{\frac{g}{l}} \stackrel{\text{electromagnetic}}{=} \dots = \sqrt{\frac{1}{LC}} \stackrel{QM}{=} \frac{1}{2} \frac{p^2}{m \hbar} \quad (1)$$

$$\nu = \frac{E}{\hbar} = \frac{v_{phs}}{\lambda} = \frac{c^2}{v} \frac{1}{\lambda} = \frac{\omega}{k} \frac{1}{\lambda} = \frac{\omega}{2\pi} \quad (2)$$

$$\nu_i = \frac{eV_{0i}}{\hbar} + \nu_0 = \frac{E_{kin}}{\hbar} \omega \mathcal{T} + \nu_0 \quad (3)$$

$$e = \text{electron charge} \quad V_0 = \text{Opposing Potential } (I = 0) \quad (4)$$

Energy Level: $i = [1, n] \in \mathbb{N}$

$$E_{kin}^{max} = eV_0 = \hbar(\nu - \nu_0) \quad (5)$$

Frequency contextual Causes

Frequency of rotational velocity as periodic occurance of an event/state by space distance and time periodicity as well as movement speed and reach length, $v^2 = v_{ph} v_{gr}$, $\omega t = 2\pi \nu t = \lambda \neq 0$:

$$\frac{v}{r} = \omega = \frac{\lambda}{t} \quad (6)$$

Rest state of source charge

Statics localization of charge. Only spacial demependency, no time dependency.

Energy transport ... Information transmission ...

$$v = 0 \quad \dot{v} = 0$$

Slow Movement of source charge

Stationarity localization of current flow by slow movement of charge. Dynamics, time dependency.

Energy transport ... Information transmission ...

$$v = \text{const.} \ll c \quad \dot{v} = 0$$

Fast Movement of source charge

Fields forces by fast movement of charge. Dynamics, time dependency.

Energy transport ... Information transmission ...

$$v \approx c \quad \dot{v} = 0$$

Acceleration Movement of source charge

Radiation energy by accelerated movement of charge. Dynamics, time dependency.

Energy transport ... Information transmission ...

$$v \neq 0 \quad \dot{v} \neq 0$$

Movement in Vacuum

$$\eta = \frac{\tan \alpha}{\tan \beta} = \frac{c}{v_{ph}} = 1 \quad 2\pi = k\lambda \quad v_{ph} = c \quad \omega = kc$$

$$\varepsilon_0 \mu_0 = \frac{1}{c^2}$$

$$\text{Continuity: } \nabla \cdot j = i\omega \varrho \quad i\omega Q = 0$$

Movement in Matter

$$\eta = \frac{\tan \alpha}{\tan \beta} = \frac{c}{v_{ph}} = \sqrt{\frac{\varepsilon\mu}{\varepsilon_0\mu_0}} \neq 1 \quad v_{ph} = \pm \sqrt{\frac{1}{\varepsilon\mu} - \frac{\sigma^2}{4\varepsilon^2 k^2}} \quad \omega = kv_{ph} = kc \sqrt{\frac{\varepsilon_0\mu_0}{\varepsilon\mu}} =$$
$$kc \frac{\tan \beta}{\tan \alpha} = k \frac{c}{\eta}$$

$$\varepsilon\mu = \left(\frac{\eta}{c}\right)^2$$

$$\text{Dispersion: } k^2 = \varepsilon\mu\omega^2 + i\mu\sigma\omega$$

Near Field

$$r_0 \ll r \ll \lambda \quad kr \ll 2\pi \quad \frac{w_m}{w_e} \ll 1$$

r is small, fast movement

Far Field

$$r \gg \lambda \quad kr \gg 1 \quad r \gg kr_0^2 \ll 1 \quad \frac{w_m}{w_e} = 1$$

r is big, slow movement

High Frequency

$$\omega \gg \frac{2}{\mu\sigma r_0^2}$$

λ is big, ν frequency is small, short time period

Low Frequency

$$\omega \ll \frac{2}{\mu\sigma r_0^2}$$

λ is small, ν frequency is big, long time period

Conditions

Border (skin surface): ...

Interior (medium body): ...

Global:

Local:

Consequences

dielectricity constant (matter medium): $\varepsilon = \varepsilon(\omega) = \mathcal{E}^{-1} \mathcal{D} = \mathcal{E}^{-1} (\mathcal{P} + \varepsilon_0 \mathcal{B})$

permeability constant (matter medium): $\mu = \mu(\omega) = \mathcal{B}^{-1} \mathcal{H} = \mathcal{B}^{-1} \left(\frac{1}{\mu_0} \mathcal{B} - \mathcal{M} \right)$

conductivity (matter medium): $\sigma = \sigma(\omega) = \frac{\sigma_0}{\sqrt{1 + \omega^2 t^2}}$

magn. energy density (vacuum): $w_m = \frac{1}{2} \frac{1}{\mu_0} \mathcal{B}^2$

elec. energy density (vacuum): $w_e = \frac{1}{2} \varepsilon_0 \mathcal{E}^2$

wave vector: $k = \frac{2\pi}{\lambda} = \eta \frac{\omega}{c}$

refraction index (matter medium): $\eta = \frac{\tan \alpha}{\tan \beta} = \frac{\eta_{medium1}}{\eta_{medium2}}$

Energy: $E = h\nu = \hbar\omega = E_{kin} + E_{pot} \stackrel{!}{=} \sqrt{(pc)^2 + (mc^2)^2}$